

React Fundamentals

Building Professional-Grade Interfaces for Zindua Market

Module 01: Introduction

What is React?

A Modern Library

- Created by Meta (Facebook) for building UIs.
- **Component-Based:** Breaking the UI into small, isolated pieces.
- **Virtual DOM:** An efficient way to update the browser only when necessary.
- Industry standard for scalable applications.



Why the Industry Prefers React

Declarative UI

Describe what you want to see, and React takes care of the "how" (DOM updates). This leads to more predictable and easier-to-debug code.

Reusability

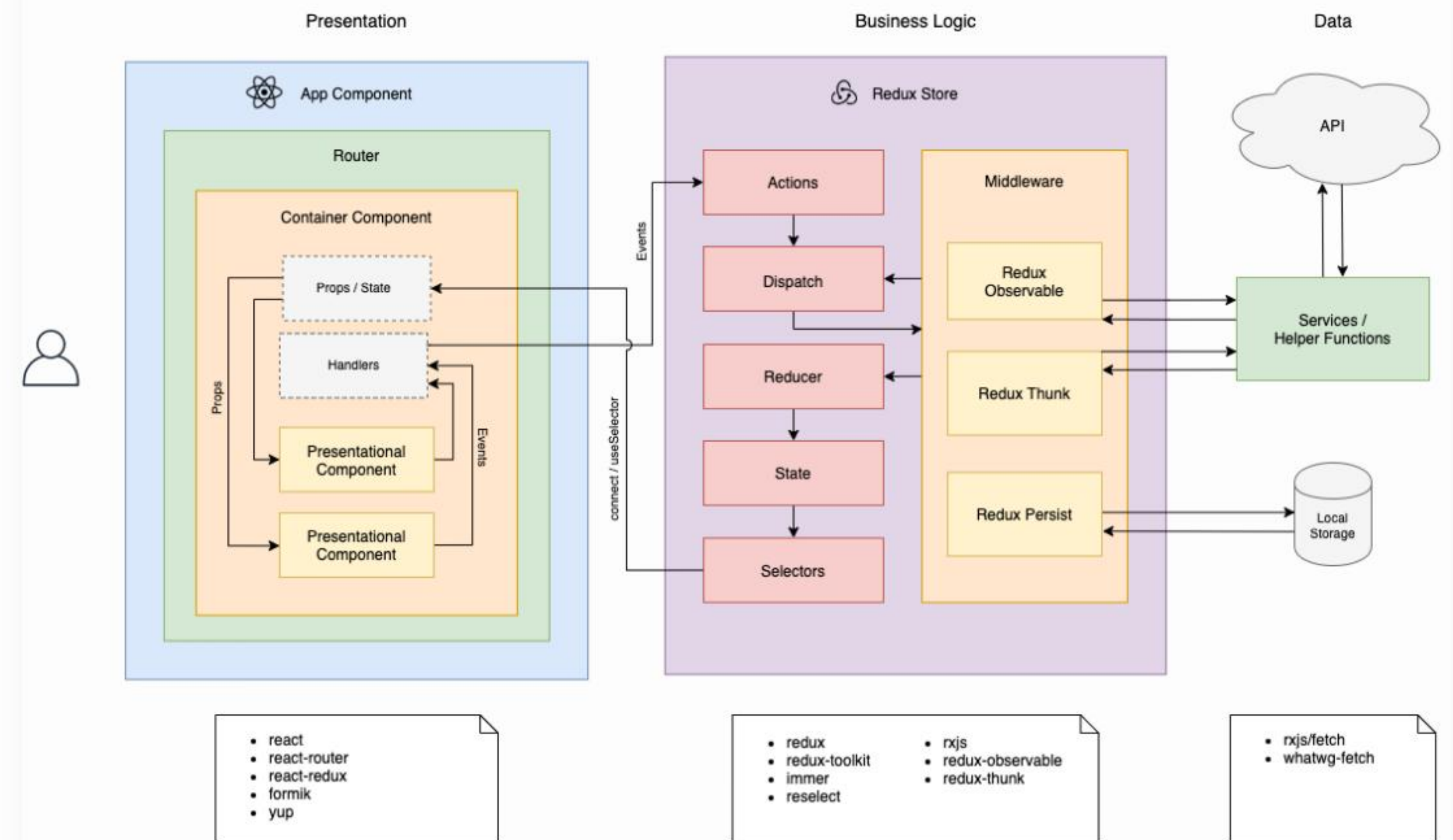
Build a component once (like a button or navbar) and use it across your entire Zindua Market application without duplication.

Module 02: JSX Explained

The Power of JSX

JavaScript XML

- A syntax extension that looks like HTML but lives in JS.
- Allows you to write **UI Logic** and **Markup** in one place.
- Embed dynamic data using { curly braces }.
- Transpiled to regular JS for browser compatibility.



Unmatched Performance

60

FPS Rendering

Virtual DOM Optimization

React creates a lightweight copy of the DOM. When state changes, it "diffs" the copy against the original and updates **only** the modified parts, ensuring smooth interactions even in complex apps.

Module 03: Components

The Building Blocks of UI



Independent

Each component manages its own logic, making it easy to test and maintain.



Reusable

Create a single "Product Card" and render it hundreds of times with different data.



Hierarchical

Complex UIs are built by nesting smaller components within larger ones.

Module 04: Imports & Exports

Organizing Your Project

Default Export

Used for the main component in a file. You can only have **one** default export per module. Import as: `import Header from './Header'`.

Named Exports

Used when exporting multiple items (like utility functions or icons) from one file. Import using braces: `import { Button, Input } from './UI'`.

Module 05: Data Flow (Props)

Passing Data as Props

Unidirectional Flow

- **Props** are "properties" passed from Parent to Child.
- They are **Read-Only** (Immutable) to the child component.
- Think of them as function arguments for your UI components.
- This ensures data flow is predictable and easy to track.

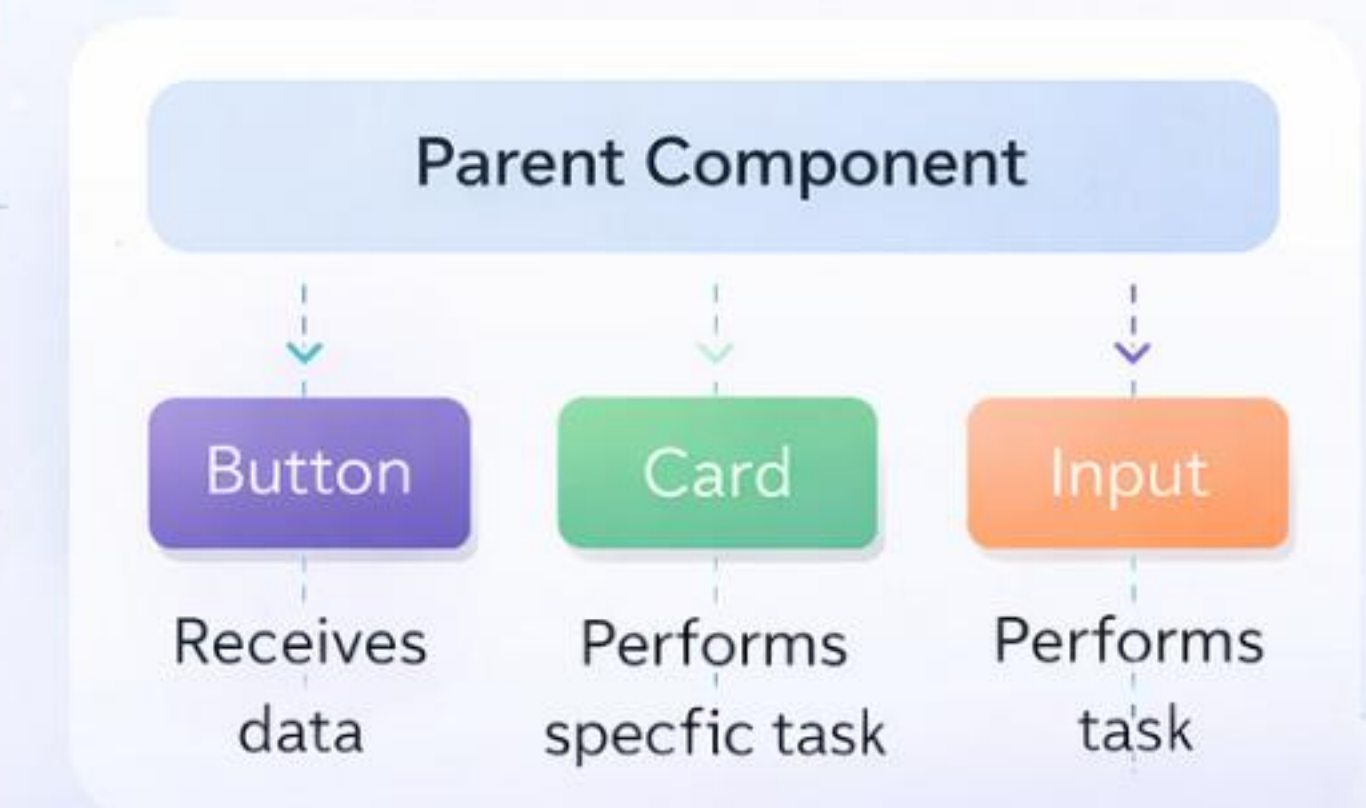
Parent & Child Components

Understanding UI Structure Through Relationships



- ➡ Sets rules & structure
- ➡ Passes data

- Passes data
- Receives data



Questions?

Thank you for your attention. Let's build Zindua Market!

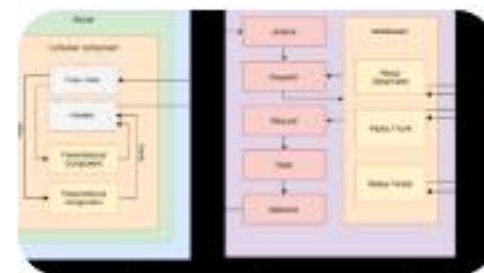
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